

Internship Report

AWS in a Hospital Management System

Introduction

Today's internship session was about Amazon Web Services (AWS). Instead of learning only the theoretical concepts of AWS services, I tried to understand how they can be applied in a real-world project. I selected a **Hospital Management System** because hospitals need to manage patient records, doctor appointments, medical reports, and billing in a secure and efficient manner. AWS provides cloud services that make the system scalable, reliable, and easy to maintain.

Why Cloud Computing?

If a hospital manages its application using its own servers, it must take care of hardware, software updates, backups, storage, and security. These tasks require significant time and cost. By using AWS cloud services, the hospital can focus on providing better healthcare while AWS handles the infrastructure. The hospital can also increase or decrease computing resources based on the number of users and pay only for the resources it uses.

Amazon EC2

Amazon EC2 can be used to host the Hospital Management System. Doctors, patients, receptionists, and administrators can access the application through the EC2 server. During emergencies or peak hospital hours, if many users access the system simultaneously, the EC2 instance can be upgraded easily without affecting the application. This ensures better performance and availability.

AWS Lambda

AWS Lambda is useful for performing automatic background tasks. For example, when a patient books an appointment, Lambda can automatically send a confirmation email or SMS. It can also remind patients about upcoming appointments, notify doctors about schedule changes, and generate alerts for emergency cases. Since Lambda runs only when required, it helps reduce operational costs.

Amazon S3

Hospitals generate many digital files such as medical reports, prescriptions, X-ray images, MRI scans, and laboratory reports. These files can be securely stored in Amazon S3. Storing documents in S3 improves storage management, provides high durability, and allows authorized users to access reports whenever required. Older medical records can also be moved to lower-cost storage classes.

Database Services

Patient information, doctor details, appointment schedules, billing records, prescriptions, and treatment history can be stored in Amazon RDS. If the hospital needs very fast access to frequently used patient data, DynamoDB can also be considered. Managed database services reduce maintenance work, improve reliability, and ensure data availability.

IAM and Security

AWS Identity and Access Management (IAM) provides secure access control for different users. Doctors can access patient medical records, receptionists can manage appointments, administrators can manage the entire system, and patients can view only their own reports. This role-based access control protects confidential medical information and improves system security.

Load Balancer and Reliability

Large hospitals may have thousands of users accessing the system every day. An AWS Load Balancer distributes incoming requests across multiple EC2 instances to maintain smooth performance. AWS monitoring and backup services also help prevent downtime and ensure that important medical data remains safe and available even during unexpected failures.

Conclusion

From today's internship session, I understood that AWS is not only suitable for large technology companies but also for healthcare applications. Services such as EC2, Lambda, S3, RDS, IAM, and Load Balancer can work together to build a secure, scalable, and reliable Hospital Management System. This practical example helped me understand how AWS services solve real-world healthcare challenges instead of simply learning their definitions.